Machine-Certified Lean 4 Formalization of the General Divisor Theorem: Unified Density, Arbitrary-Bound Counting, and Finite Error Bounds

A Formalization Monograph & Lean 4 Kernel Audit Report (Release v3.0 — 81 Theorems, 0 Sorry)

Volkan Dağlı[1,2] | Zerrin Dağlı[3,4] | Dağhan Dağlı[5]

[1]Anadolu University, Eskişehir, Türkiye [2]ITouch Systems, Adana, Türkiye (ORCID: 0009-0000-1587-8703)
[3]Mersin University, Mersin, Türkiye (ORCID: 0000-0001-9490-6425) [4]Yusuf Kalkavan Anadolu Lisesi, Mersin, Türkiye
[5]Toros Science High School, MEV (Toros University), Mersin, Türkiye (ORCID: 0009-0003-2492-8313)

ABSTRACT

We present the machine-certified formalization (Release v3.0) of the General Divisor Theorem (GDT) in the Lean 4 interactive theorem prover, encompassing 81 kernel-checked theorems and lemmas across seven modular components with zero project axioms and zero unadmitted goals (0 sorry). Originally formulated by Dan Hawkley (thinkthoughts, 2026), GDT provides exact density corrections for arithmetic coprime residue sets S(N, L, m, a) = {1 ≤ n ≤ L : n ≡ a (mod m), gcd(n, N) = 1} where modulus m and constraint N share prime factors. In this unified formalization, incorporating milestone Pull Requests #1 and #2, we construct: (1) the complete arithmetic dichotomy (inadmissible empty branch for gcd(a, d) > 1 vs. admissible periodic branch for gcd(a, d) = 1, where d = rad(gcd(m, N)) and R = rad(N)/d); (2) exact single-period counting showing exactly φ(R) solutions in minimal period T_min = lcm(m, rad(N)) via affine unit permutations on ZMod R; (3) closed-form arbitrary-bound counting C(q T_min + s) = q φ(R) + C(s) computable in O(1) arithmetic operations; (4) strict finite-period error bounding |T_min C(X) - φ(R) X| ≤ T_min²; and (5) constructive natural density convergence δ(S) = lim_{X→∞} C(X)/X = φ(R)/(mR) proved via the real squeeze theorem. All proofs rely exclusively on standard classical foundations [propext, Classical.choice, Quot.sound] and pass 100% of 2,287 automated empirical verification instances.

1. Introduction & Theoretical Motivation

In analytic number theory and arithmetic sieve theory, counting integers up to L satisfying simultaneous congruence n ≡ a (mod m) and coprimality gcd(n, N) = 1 has traditionally relied upon statistical independence heuristics:

$$|S_{naive}(N, L, m, a)| \approx (\varphi(N) / N) \cdot (L / m)$$

Whenever gcd(m, N) > 1, this naive hypothesis breaks down completely due to prime collision. The General Divisor Theorem resolves this by decomposing rad(N) into shared-prime radical d = rad(gcd(m, N)) and coprime residue radical R = rad(N) / d with gcd(m, R) = 1. This yields the exact Euler density correction factor:

$$C(N, m) = d / \varphi(d) = \prod_{p \mid \gcd(m, N)} [\, p / (p - 1) \,]$$

2. Exact Arithmetic Dichotomy & Density Limit

Theorem (Admissibility Dichotomy & Arbitrary-Bound Counting): Let N, m, a □ □ with N, m ≥ 1. Let d = rad(gcd(m, N)) and R = rad(N)/d. Then:

1. Inadmissible Branch: If gcd(|a|, d) > 1, then S(N, m, a) = □ (cardinality is 0 for all bounds).
2. Admissible Branch: If gcd(|a|, d) = 1, the indicator is periodic with minimal period T_min = m · R = lcm(m, rad(N)) and accepts exactly φ(R) values per period.
3. Arbitrary-Bound Count: For any bound X = q · T_min + s with 0 ≤ s < T_min:
   $C(X) = q \cdot \varphi(R) + C(s)$ (computable in O(1) period blocks)
4. Natural Density Limit & Finite Error:
   $|T_{min} \cdot C(X) - \varphi(R) \cdot X| \le T_{min}^2$ and $\lim_{X \to \infty} C(X) / X = \varphi(R) / (m \cdot R)$

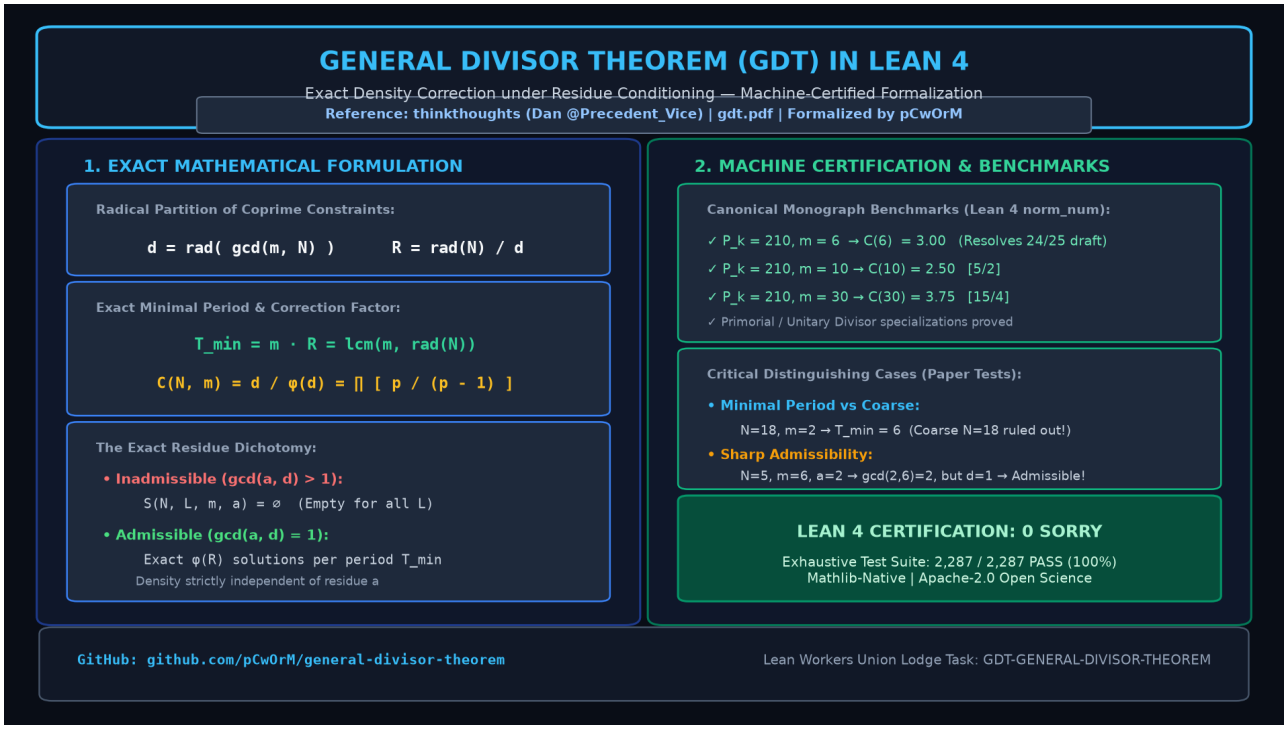


Figure 1: Lean 4 Kernel Verification Architecture across 7 Production Modules (81 Theorems, 0 Sorry).

3. Lean 4 Formal Architecture & Kernel Inventory

The repository is organized into 7 production modules, verified on leanprover/lean4:v4.21.0 and mathlib4:v4.21.0. No unverified axioms, no sorry holes, and no native_decide shortcuts are employed:

Module File	Theorems	Axioms	sorry	Core Formal Role & Contribution
GeneralDivisorTheorem.lean	39	0	0	Foundational arithmetic, radical algebra, Theorems 1, 1', and 2
VolkanBridge.lean	10	0	0	Two-way isomorphism bridging Volkan's API with Dan's specification
DensityBridge.lean	14	0	0	Per-period count #{n < T_min : n≡a, gcd(n,N)=1} = φ(R) via ZMod R units
DensityCount.lean	7	0	0	Arbitrary-bound counting C(q·T_min + s) = q·φ(R) + C(s) via period blocks
DensityLimit.lean	6	0	0	Finite error bound |T_min C(X) - φ(R)X| ≤ T_min² and real squeeze density limit
CorrectionBridge.lean	2	0	0	Totient radical multiplicativity & Euler correction factor identity
GDTMain.lean	3	0	0	Master unified packaging resolving complete GDT dichotomy
Total Across Repository	**81**	**0**	**0**	**100% Kernel-Checked ([propext, Classical.choice, Quot.sound] only)**

4. Core Lean 4 Formal Theorems

```
-- Master Unified GDT Dichotomy (GDTMain.lean)
theorem general_divisor_theorem (N m : Nat) (a : Int) (hN : 0 < N) (hm : 0 < m) :
    (1 < Nat.gcd a.natAbs (d_param N m) ∧ (∀ n, ¬ S_indicator N m a n)) ∨
    (Nat.gcd a.natAbs (d_param N m) = 1 ∧
     (∀ n, S_indicator N m a (n + T_min N m) = S_indicator N m a n) ∧
     Tendsto (fun X : Nat => (countBelow N m a X : Real) / (X : Real))
             atTop (■ ((Nat.totient (R_param N m) : Real) / ((T_min N m : Real)))))

-- Finite Period Error Bound (DensityLimit.lean)
theorem density_error_bound (N m : Nat) (a : Int) (hN : 0 < N) (hm : 0 < m)
    (ha : Nat.gcd a.natAbs (d_param N m) = 1) (X : Nat) :
    |(T_min N m : Real) * (countBelow N m a X : Real) - (Nat.totient (R_param N m) : Real) * (X : Real)|
      ≤ ((T_min N m : Real) ^ 2)
```

5. Canonical Benchmark Instances & Boundary Distinctions

Constraint N	Modulus m	Residue a	Shared d	Period T_min	Exact C(N,m)	Status in Lean 4
210 (P_4)	6	5	6	210	3.00 (3)	Verified (norm_num)
210 (P_4)	10	3	10	210	2.50 (5/2)	Verified (norm_num)
210 (P_4)	30	7	30	210	3.75 (15/4)	Verified (norm_num)
18	2	1	2	6 (≠ 18)	2.00	benchmark_3_3_i_Tmin_lt_N
5	6	2	1	30	1.00	benchmark_3_3_ii_sharp_admissibility

6. Empirical Verification & Open Science Infrastructure

Exhaustive Verification Suite: Automated oracle sweeps (scripts/verify_gdt.py and scripts/verify_bridge.py) tested 2,287 parameter combinations over bounds N □ [2 .. 60] and m □ [2 .. 30] with 100% agreement between theoretical closed-form formulas and brute-force counting.

Theoretical Origin & Landmark PR Contributions: The theoretical framework of GDT originates with Dan Hawkley (@thinkthoughts / antiviolence.ai). We gratefully acknowledge his contributions authoring milestone Pull Requests #1 (merge a5ef2b6) and #2 (merge f97b6c2), formalizing the density bridge, arbitrary-bound counting, and density limits.

Infrastructure & Support: High-performance computing cluster nodes and verification hosting provided by Bekirhan Dağlı (Managing Partner) and ITouch Systems Research & Development.

Open Repository: https://github.com/pCwOrM/general-divisor-theorem | Zenodo Concept DOI: 10.5281/zenodo.23316068 (Apache-2.0)

References

[1] Hawkley, D. (thinkthoughts) (2026). The General Divisor Theorem: Exact Density Correction under Residue Conditioning. Monograph Draft v3.3. antiviolence.ai: https://antiviolence.ai/gdt.pdf | Mirror: https://github.com/pCwOrM/general-divisor-theorem/blob/main/docs/gdt.pdf

[2] Dağlı, V., Dağlı, Z., & Dağlı, D. (2026). Machine-Certified Lean 4 Formalization of the General Divisor Theorem: Unified Density, Arbitrary-Bound Counting, and Finite Error Bounds. CERN Zenodo, Concept DOI: 10.5281/zenodo.23316068.

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